

RFID-Based Door Lock System

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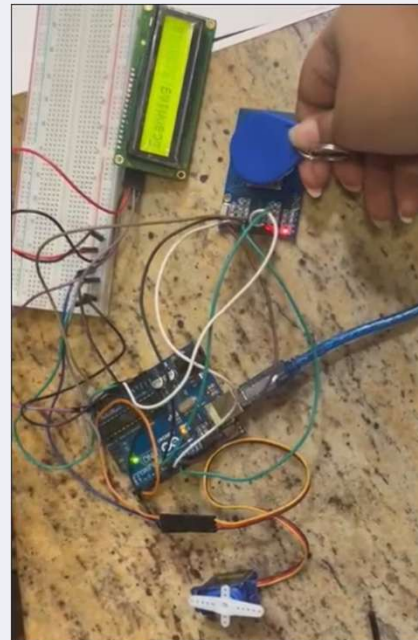
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Project Topic:

Design and implementation of an RFID-based door lock system

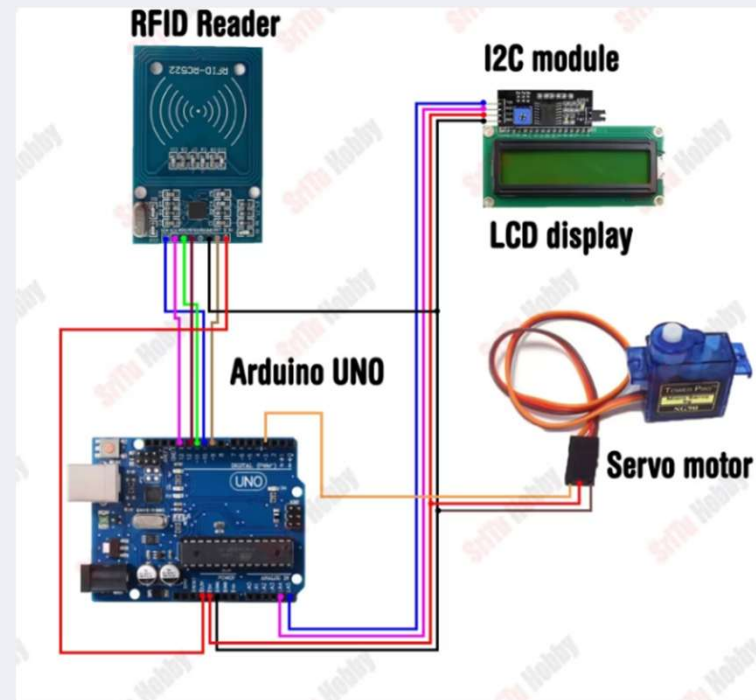
Project Interface :



INTRODUCTION

This project introduces a secure, contactless door locking system.

It utilizes an RC522 RFID Reader, Arduino Mega, Servo Motor, and I2C LCD. This provides access control without a keypad, ensuring simplicity and enhanced security for homes and businesses.



Objectives

Enhanced Security

Purpose: Enhance security with contactless technology, offering a modern approach to access control.

Cost-Efficient System

Goal: Develop a cost-efficient and scalable system, making it accessible for various applications.

Versatile Applications

Applications: Ideal for residential and commercial security systems, providing secure access.

Components and Hardware

Essential hardware components for the RFID door lock system.

1 RC522 RFID Reader

Reads unique RFID tags for secure authentication.

2 Arduino UNO

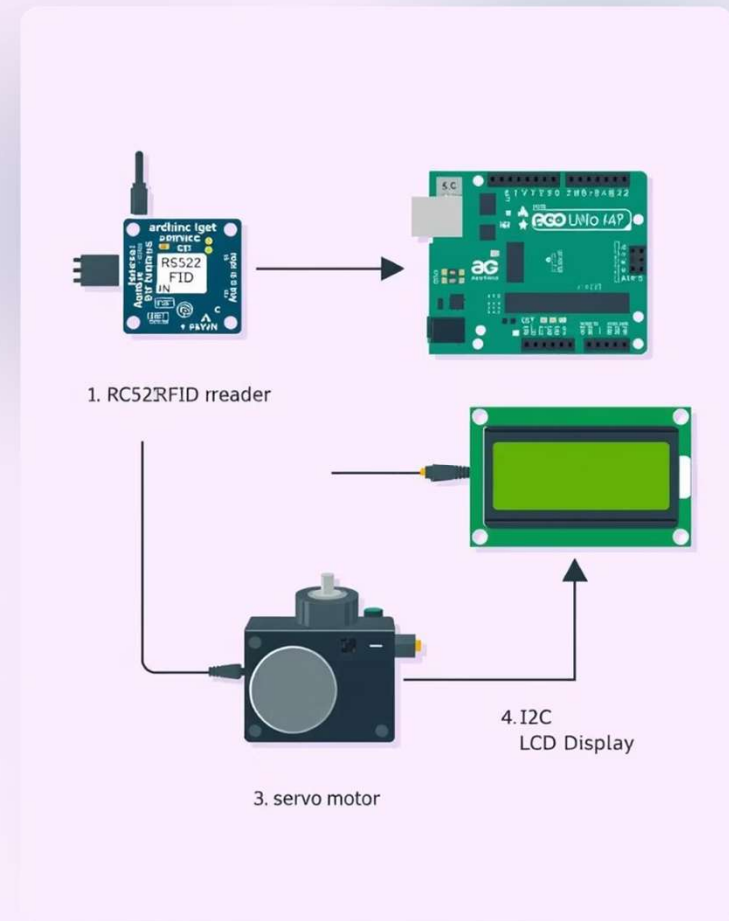
The microcontroller to process data and control the lock.

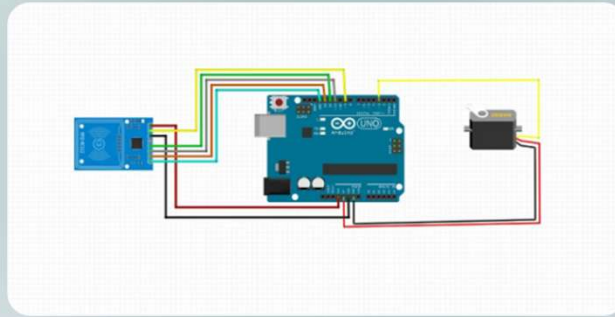
3 Servo Motor

Physically locks and unlocks the door mechanism.

4 I2C LCD

Displays system status and user feedback.





Circuit Connections

SPI Connection

RC522 connected to Arduino via SPI pins (SDA, SCK, MOSI, MISO) for data transfer.

I2C LCD Setup

I2C LCD communication setup for clear and effective display of messages.

PWM Servo Control

Servo motor connected to a digital PWM pin, enabling precise lock control.

Stable Power

Powered with 5V adapters ensuring stable operation of all components.

System Workflow



Display Message

LCD displays "Scan Your Card" prompting user interaction.



Retrieve UID

RFID reader retrieves UID from the scanned card for verification.



Verify UID

Arduino verifies the UID against a predefined authorized database.



Access Status

Displays "Access Granted" or "Access Denied" based on UID verification.



Actuate Lock

Servo motor locks or unlocks the door as needed to grant access.

Key Features

1 RFID Verification

Secure access through RFID verification using the MFRC522 library.

2 Feedback Messages

Real-time feedback via the LiquidCrystal_I2C library for enhanced UX.

3 Authorized UID List

Predefined list ensures only permitted users can gain access.

4 Servo Control

Precise servo logic manages secure lock actuation.





Challenges

RFID Reader Issues

Reliable tag detection, interference from metal/RF signals.

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Limited RFID Range

Short 2-3 cm range, inconvenient for users.

Servo Power Fluctuations

Unlocking draws more power, can reset Arduino.

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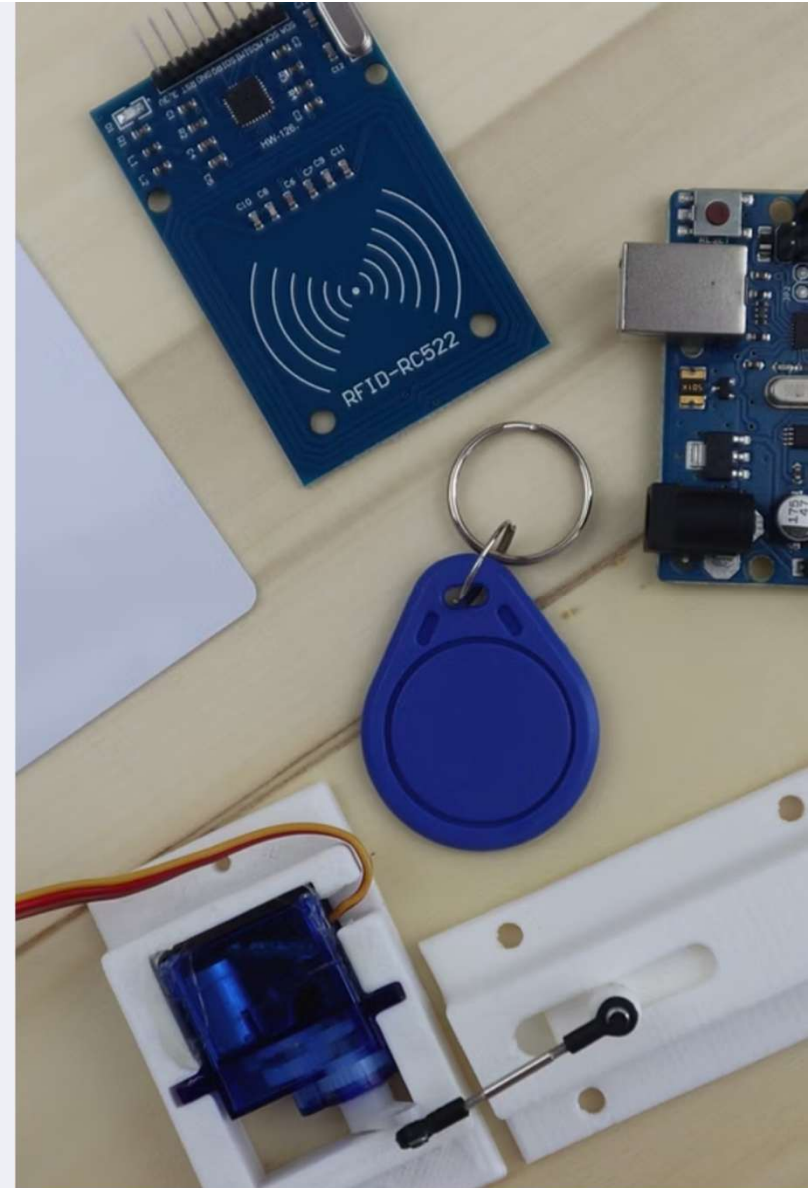
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Security Limitations

Low-frequency RFID cards can be cloned.

Conclusion

As we wrap up our exploration of the RFID-based door lock system, we've delved into its components, circuit connections, code features, and challenges. The system's scalability and potential for future upgrades make it a promising solution for secure access control.



Upgrades

1 Highlights

RFID-based smart lock using Arduino for secure, contactless access.

3 Future Additions

Biometric login or SMS/WhatsApp alerts can enhance security.

2 Scalability

Can connect with mobile apps or cloud for advanced control.

4 IoT Potential

Great example of Arduino's role in smart, connected systems.